x	-1	0	1
y	-4	-1	2

The points (x,y) to be plotted :

$(-1, -4)$, $(0, -1)$ and $(1, 2)$.

(ii) Let us prepare a table to find the ordered pairs of points

for the line $y = \left(\frac{2}{3}\right)x + 3$.

Let us assume -3, 0, 3 as x values. (why?)

When $x = -3$, $y = \frac{2}{3}(-3) + 3 = 1$

When $x = 0$, $y = \frac{2}{3}(0) + 3 = 3$

When $x = 3$, $y = \frac{2}{3}(3) + 3 = 5$

x	-3	0	3
y	1	3	5

The points (x,y) to be plotted :

$(-3,1)$, $(0,3)$ and $(3,5)$.

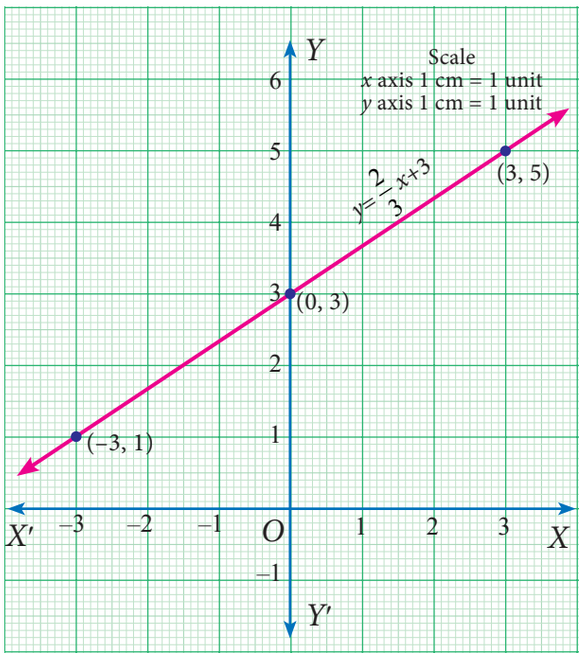


Fig. 3.17

3.8.1 Simultaneous Linear Equations

With sufficient background of graphing an equation, now we are set to study about system of equations, particularly pairs of simultaneous equations.

What are simultaneous linear equations? These consists of two or more linear equations with the same variables.

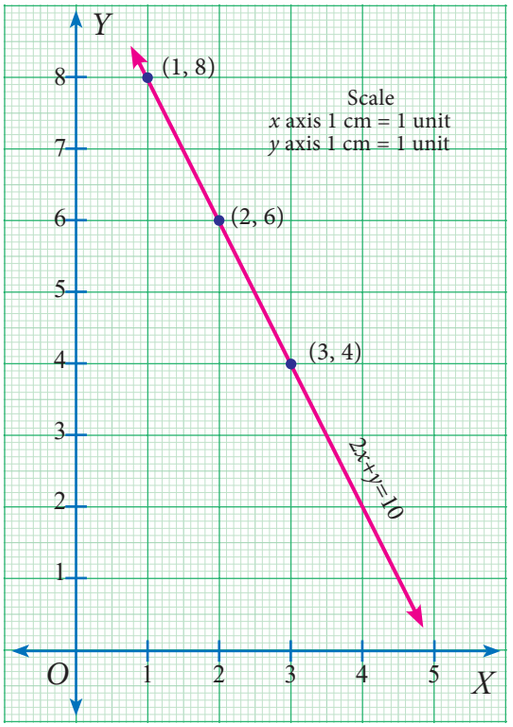


Fig. 3.18

Why do we need them? A single equation like $2x+y=10$ has an unlimited number of solutions. The points (1,8), (2,6), (3,4) and many more lie on the graph of the equation, which means these are some of its endless list of solutions. To be able to solve an equation like this, another equation needs to be used alongside it; then it is possible to find a single ordered pair that solves both equations at the same time.

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